MIL-T-82612A(OS)
31 January 1975
Superseding
MIL-T-82612(OS)
3 August 1972

MILITARY SPECIFICATION

TORPEDO TUBE, SURFACE VESSEL, MARK 32 MOD 11

(For Torpedoes Mark 44 Mod 1 and Mark 46 Mod 1)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 This specification establishes the requirements for the procurement of Torpedo Tube Assembly Mk 32 Mod 11 (LD 632038).

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposals form a part of this specification to the extent specified herein. In the event of conflict between this specification and other documents referenced herein, the requirements of this specification shall apply.

SPECIFICATIONS

Military

MIL-C-104	Crates, Wood; Lumber, and Plywood; Sheathed, Nailed, and Bolted
MIL-P-116	Preservation - Packaging, Methods of
MIL-S-901	Shock Tests, H. I. (High-Impact); Shipboard Machinery, Equipment and Systems; Requirements for
MIL-C-16173	Corrosion Preventive Compound, Solvent Cutback Cold Application

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

FSC 1045

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

MIL-STD-167	Mechanical Vibrations of Shipboard Equipment
MIL-STD-810	Environmental Test Methods

DRAWINGS

Naval Sea Systems Command (Code Ident 10001)

LD 162748	Torpedo Tube, Surface Vessel, Mk 32 Mod 11 Assembly
LD 632038	Torpedo Tube, Surface Vessel, Mk 32 Mod 11 Index to List of Drawings
Dwg 1623864	Power Unit, Hydraulic
Dwg 1623922	Pressure Gage
Dwg 1984119	Dummy Torpedo Mk 44 Mod 1
Dwg 2320787	Barrel Assembly Gage "GO" Torpedo Tube Mk 32 Type
Dwg 2652289	Dummy Torpedo Mk 46 Mod 1
Dwg 3014877	External Electrical Schematic
Dwg 3015069	Wiring Diagram
Dwg 3245120	Pressure Fitting Adapter

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

National Bureau of Standards Publication

Handbook H28	Screw Thread Standards for the Federal Services

(Application for copies should be addressed to the superintendent of Documents, Government Printing Office, Washington, DC 20402.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Description. Torpedo Tube Assembly Mk 32 Mod 11 is a pneumatic launching device capable of receiving, stowing, and launching Torpedoes Mk 44 Mod 1 and Mk 46 Mod 1. The tube assembly consists of a fiberglass reinforced Torpedo Tube Barrel Mk 10 Mod 19 fitted with Breech Mechanism Mk 1 Mod 1 and a torpedo securing mechanism. The barrel can be fired electrically from a remote location or manually at the tube in an emergency condition. During manual firing, a latch plate is swung open to (1) provide automatic switching from normal to emergency torpedo gyro squib electrical power and (2) expose a firing button for actuating the solenoid valve manually. The tube is constructed with a junction box that receives electrical functions from the fire control system and transmits them via two electrical cables to the control box and torpedo preset cable inside the breech access housing. A charging line stowage tube is mounted on the tube to accommodate the flexible charging hose.

3.2 General requirements

3.2.1 Preproduction sample. Unless otherwise specified in the contract or order, a preproduction sample of the torpedo tube assembly shall be manufactured using the methods and procedures proposed for the production lot. The sample will be tested as specified in Section 4 herein and is for the purpose of determining, prior to starting production, that the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract. (See 6.2)

* 3.2.2 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of Torpedo Tube Mk 32 Mod 11 shall conform to the detailed requirements given on the drawings and specifications listed on LD 162748.

* 3.2.3. Finishes, coatings, construction, and dimensions. Finishes, coatings, construction, and dimensions shall be in accordance with the applicable drawings, specifications, and publications listed in LD 162748.

* 3.2.4 Moisture, fungus, and corrosion. The tubes shall be rendered moisture, fungus, and corrosion resistant in accordance with detail requirements specified on drawings and specifications listed in LD 162748.

* 3.2.5 Marking. Marking of assemblies, subassemblies, and parts shall be in accordance with the applicable drawings listed in LD 162748.

3.2.6 Threads. Unless otherwise specified, all threads shall be in accordance with National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

3.2.7 Safety precautions. All standard safety precautions for electrical apparatus and high pressure air shall be strictly observed. No test shall be permitted to continue if in the opinion of the test engineer it is deemed unsafe. While conducting any of the firings, personnel should stand clear of the breechward access opening and breech mechanism. Charging lines should be secured with safety chains when high pressure air charging operations are in progress.

3.2.8 Definitions. For the purpose of this specification the definitions listed in 6.3 are applicable.

3.3 Performance requirements and product characteristics. Torpedo Tubes Mk 32 Mod 11 shall meet the following performance requirements and product characteristics:

* 3.3.1 Hydrostatic pressure. The breech mechanism shall withstand a hydrostatic pressure of 4500 plus 20 minus 0 psig without failure of any part. The low-pressure pneumatic system of the torpedo tube assembly shall withstand a hydrostatic pressure of 225 plus 20 minus 0 psig without failure of any component in the pneumatic system.

* 3.3.2 Pneumatic system pressure drop. The high-pressure pneumatic system of the torpedo tube, minus the breech mechanism, shall be attached to a 2000 psi air source through a 0.024 inch diameter orifice. With the securing mechanism removed and the manifold nipple blanked off, the pressure in the pneumatic system of each barrel shall stabilize at 1350 psig minimum. With the securing mechanism installed, the pressure shall stabilize at 750 psig minimum. At the completion of this test there shall be no deformation or failure of any part of the pneumatic system.

3.3.3 Breech mechanism air loss. After being initially charged to a pressure of 2000 plus or minus 50 psig, allowed to cool for 2 hours, and then bled down to 1600 plus or minus 10 psig, the breech mechanism shall not lose more than 5 percent air pressure, due to leakage, over a 24-hour period.

* 3.3.4 Bore size of barrel. Each barrel shall freely pass a "GO" gage (Dwg 2320787) with an axial force not to exceed 150 pounds.

3.3.5 Dielectric strength. Electrical circuitry of the tube shall withstand the application of a potential of 1000 volts ac between mutually insulated terminals and ground for a minimum of one minute without breakdown.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

3.3.6 Insulation resistance. Immediately following the dielectric strength test, the insulation resistance between hipotted points shall be 3 megohms minimum when a potential of 500 volts dc is applied.

3.3.7 Circuitry. Electrical circuitry shall be in accordance with Dwg 3015069.

* 3.3.8 Barrel heater resistance. The total electrical resistance of the heater circuit shall 11.4 ohms plus or minus 0.8 ohm.

* 3.3.9 Barrel heater circuit operating current. When the barrel heater circuit is operating, the total maximum current drain shall be 11 amperes with an input of 115 volts ac plus or minus 0.5 volt.

3.3.10 Squib fire switch. The plunger shall move out forcefully when the latch plate is released.

3.3.11 Muzzle cover seal test. The axial force required to remove the muzzle cover shall be 30 pounds plus or minus 10 pounds.

* 3.3.12 Pressure switch operation. The pressure switch contacts shall close at 1275 psig plus or minus 75 psig and remain closed at pressures above this value. The pressure switch contacts shall open at 1200 psig minimum on falling pressure and remain open at pressures below this value.

3.3.13 Breakaway plug puller mechanism

3.3.13.1 The lever of the breakaway plug puller mechanism shall be capable of being latched and unlatched by hand.

3.3.13.2 The breakaway plug puller mechanism shall permit dry firings when in the latched position and shall remain in the latched position at the completion of firings.

* 3.3.13.3 The breakaway plug, during torpedo firings, shall pull away from the torpedo receptacle in Torpedoes Mk 44 Mod 1 and Mk 46 Mod 1 without damage and shall not affect torpedo exit velocities when the tube is fired at pressures of 1200 through 1500 psig.

3.3.14 Control valve operating lever. The tube shall not be capable of being fired, either electrically or manually, until the control valve operating lever is placed in the READY condition.

3.3.15 Solenoid voltage. The solenoid firing valve shall operate throughout an input voltage range of 102 through 124 volts ac, 60 Hertz (Hz), single phase.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

* 3.3.16 Manual firing. The axial force that must be applied on the manual firing button to initiate the firing cycle at pressures of 1200 psig through 2000 psig shall not exceed 25 pounds.

* 3.3.17 Dry firing test. The tube shall be capable of being fired, without a torpedo, by mechanical means at any flask pressure from 1200 to 2000 psig and by electrical means when the solenoid valve is energized for 400 milliseconds maximum at any flask pressure of 1350 through 2000 psig without damage, malfunction, or hindrance to subsequent firings with a torpedo. During dry firings of the tube and firings of dummy Torpedoes Mk 44 and Mk 46 at pressures of 1200 through 1500 psig, the plug puller mechanism shall move automatically from the unlatched to the latched position. Upon completion of firings, the lever shall be capable of being unlatched by hand.

* 3.3.18 Torpedo exit velocities. The exit velocities of dummy Torpedoes Mk 44 Mod 1 (433 plus or minus 7 pounds) and Mk 46 Mod 1 (508 plus or minus 8 pounds) shall be in accordance with Schedule II as specified in 4.4.4.17.

3.3.19 Electrical continuity. Electrical continuity shall be in accordance with Dwg 3015069.

* 3.3.20 Lanyard retainer. The spring-loaded retainer shall be capable of being depressed by hand and shall be capable of accepting and securing a lanyard wire of 0.069-inch minimum diameter.

3.3.21 Ready cylinder assembly solenoid valve. The solenoid valve shall operate throughout an input voltage range of 102 through 124 vac, 60 Hz, single phase.

3.4 Environmental requirements

3.4.1 Vibration

3.4.1.1 Vibration, nonoperating. The torpedo tube assembly, less packing or packaging, shall be capable of meeting all quality conformance tests after being subjected to vibration tests of MIL-STD-167 for Type I vibrations in the three mutually perpendicular axes.

3.4.1.2 Vibration, standby condition. The torpedo tube assembly, loaded with a dummy Torpedo Mk 46 Mod 1 and containing a breech assembly pressurized to 1500 psig, shall be capable of meeting all quality conformance tests during and after being subjected to the vibration tests of MIL-STD-167 for Type I vibrations in the three mutually perpendicular axes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

3.4.2 Shock

3.4.2.1 Shock, nonoperating. The torpedo tube assembly, less packing or packaging, shall be capable of meeting all quality conformance tests after being subjected to the shock impulses of the magnitude and number specified in MIL-S-901 for medium weight equipment

3.4.2.2 Shock, loaded condition. The torpedo tube, loaded with a dummy Torpedo Mk 46 Mod 1, shall be capable of meeting all quality conformance tests after being subjected to the shock impulses of the magnitude and number specified in MIL-S-901 for medium weight equipment.

3.4.3 Temperature and humidity

* 3.4.3.1 Low temperature operation, loaded condition. The torpedo tube, less packaging and packing and in the loaded condition, shall be capable of withstanding the low temperature conditions specified in 4.5.3.1. While at the low test temperature, the barrel heaters shall maintain the temperature on the outside surface of the torpedo at not less than plus 33 degrees Fahrenheit (F). At the conclusion of the test and while still at the test temperature, the torpedo tube shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

* 3.4.3.2. High temperature operation. The torpedo tube, with a fully pressurized breech assembly, shall be capable of withstanding the high temperature and humidity conditions specified in 4.5.3.2. While at the high temperature and humidity conditions, the torpedo tube shall meet all the performance requirements and product characteristics of this specification. The barrel heater relay shall open the heater circuit during high temperature conditions and close it during low temperature portions of all cycles. At the conclusion of the test the torpedo tube shall show no corrosion or damage.

* 3.4.4 Marine atmosphere environment. The torpedo tube shall be capable of withstanding the marine atmosphere or alternate test specified in 4.5.4 and thereafter shall show no evidence of corrosion or damage and shall meet all the performance requirements and product characteristics of this specification.

3.5 Cleanup. Prior to and after final assembly, all parts and components shall be throughly cleaned of all loose, spattered, or excess solder, metal chips, and other foreign material. All burrs and sharp edges as well as flashing shall be removed.

3.6 Workmanship. The equipment, including all parts and accessories, shall be constructed and finished in a manner to ensure compliance with all requirements of this specification. Particular attention shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, riveting, machine screw assemblage, welding, brazing, and freedom of parts from burrs and sharp edges. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The examination and testing of Torpedo Tube Mk 32 Mod 11 shall be classified as follows:

* 4.2.1 Preproduction and periodic production tests. Preproduction and periodic production tests are those which are accomplished on samples selected as specified in 4.3.1 and 4.3.3, which are representative of the production of the torpedo tube assembly after the award of the contract, to determine that the lot and production meet the requirements of this specification. These tests are detailed in 4.5. Acceptance shall be based on no defects in a sample. Failure of a sample to comply with these requirements will result in the rejection of the lot, as determined by the procuring activity.

4.2.2 Quality conformance tests. Quality conformance tests are those performed on each torpedo tube assembly to be submitted for acceptance under contract. These tests are detailed in 4.4.

4.3 Sampling

4.3.1 Preproduction sample. Unless otherwise specified for each contract or order, a representative sample of one torpedo tube assembly manufactured in accordance with 3.2.1 shall be subjected to the preproduction tests detailed in 4.5 at an activity designated by the procuring activity. Further production of the torpedo tube by the contractor prior to approval by the contracting activity of the preproduction sample shall be at the contractor's risk. Preproduction samples accepted will be applied as part of the quantity specified in the contract or order (See 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

4.3.2 Lot size. Unless otherwise specified in the contract or order, each shipment of units of product shall be a separate "inspection lot" and may differ from the quantity specified in the total order or contract. In any event, all units of a single "inspection lot" shall be made to the same design, with the same materials and manufacturing techniques, and shall be part of one continuous production run.

4.3.3 Periodic production sample. From each succeeding lot of 50 or less that have satisfactorily passed the quality conformance inspection of 4.4, a random sample of one torpedo tube assembly shall be selected for submission to an activity designated by the procuring activity to determine compliance of the sample with the requirements of the contract, specification, and drawings.

4.3.4 Quality conformance sampling. The tests and examinations detailed in 4.4 shall be conducted on each torpedo tube assembly to be delivered; therefore, there shall be no sampling for quality conformance. Defective units shall be rejected.

4.4 Quality conformance inspection. Each Torpedo Tube Mk 32 Mod 11 to be delivered under contract shall be subjected to the tests and examinations specified under this section.

4.4.1 Test conditions. Unless otherwise specified, the torpedo tube shall be subjected to quality conformance tests under the following conditions:

4.4.1.1 Test sequence. The sequence of testing procedures presented herein is suggested only. The tests may be performed in any order that will permit the most practical and economical use of manpower and equipment.

4.4.1.2 Temperature: Room ambient 65 degrees Fahrenheit (F) to 95 degrees F (18 degrees centigrade (C) to 35 degrees C).

4.4.1.3 Altitude: Normal ground atmospheric pressure.

4.4.1.4 Vibration: None.

4.4.1.5 Humidity: Room ambient to 95 percent relative maximum.

* 4.4.2 Test results. When specified in the contract or order (See 6.2), test results shall be recorded, or verification noted, as required, opposite corresponding paragraph numbers on the Quality Conformance Inspection Data Sheet (Appendix) for each torpedo tube assembly inspected (See 5.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

4.4.3 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities and shall provide personnel for performing all quality conformance tests. The test equipment shall be adequate in quantity and, when definite requirements are not specified, shall be of sufficient accuracy and quality to permit performance of the required quality conformance tests. To perform the tests, the following items of test equipment, or equivalent, are required:

4.4.3.1 A hydraulic power unit capable of producing a hydrostatic pressure of 4500 psig (Dwg 1623864 (FSN 2A 1045-494-8324)).

4.4.3.2 A water pressure gage capable of reading 4500 psig with an accuracy of plus or minus .25 percent.

* 4.4.3.3 A 0-3000 psig high pressure air source.

4.4.3.4 An air line filter to remove water, oil, and dirt; the filter shall not pass particles larger than 50 microns.

* 4.4.3.5 An air pressure gage of 0 to at least 3000 psig with accuracy of plus or minus one-quarter percent at 2500 psig.

4.4.3.6 A "GO" bore gage in accordance with Dwg 2320787.

4.4.3.7 Standard high potential test equipment capable of an output of 1000 volts, 60 to 100 Hz.

4.4.3.8 Megohm bridge, 500 volts dc, with a range of 0.1 to 1,000,000 megohms.

* 4.4.3.9 An ohmmeter capable of measuring 11.4 ohms with an accuracy of plus or minus 5 percent of the reading.

* 4.4.3.10 A power supply capable of supplying at least 11 amperes at 115 plus or minus 0.5 volts ac, 60 Hz, single phase; the voltage must be capable of being varied from at least 102 to 124 volts ac.

* 4.4.3.11 An ac ammeter capable of measuring a current of 11 amps with an accuracy of at least plus or minus 0.375 ampere.

* 4.4.3.12 An ac voltmeter capable of measuring 102.5, 115, and 123.5 volts ac with an accuracy of at least plus or minus 1.125 volts.

* 4.4.3.13 A CO_2 bottle of sufficient volume to cool the thermoswitches for the performance of the barrel heater resistance tests and the heater circuits operating current tests.

4.4.3.14 A push scale with a memory pointer and a capacity of at least 30 pounds, with an accuracy of plus or minus 10 percent of the reading.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

* 4.4.3.15 A pull scale with a memory pointer and a capacity of at least 155 pounds, with a maximum error of plus or minus 10 percent of the reading.

* 4.4.3.16 A test circuit capable of supplying 102 to 124 volts, 60 Hz ac from the power supply of 4.4.3.10 to the solenoid for a period of 400 milliseconds maximum. Current capacity shall be at least 2 amperes.

* 4.4.3.17 Dummy Torpedo Mk 44 Mod 1 (Dwg 1984119).

4.4.3.18 Dummy Torpedo Mk 46 Mod 1 (Dwg 2652289).

4.4.3.19 Launching ramp or track on which dummy torpedoes can be launched. The ramp or track shall include a means for decelerating and stopping the dummy torpedo on the track without damage.

4.4.3.20 Two infrared photoelectric control units suitable for use with electronic counter 4.4.3.21.

* 4.4.3.21 An electronic counter capable of recording up to 10,000 milliseconds in one millisecond intervals with instant reset capabilities.

4.4.3.22 A continuity indicator.

* 4.4.3.23 A 125 plus or minus 2 volts dc power supply with a 2.7 plus or minus 0.1 K ohm series resistor.

* 4.4.3.24 Pressure fitting adapter (Dwg 3245120).

* 4.4.3.25 Nitrogen bottle of sufficient volume to purge the breech mechanism air flasks.

* 4.4.3.26 A 0.024 plus or minus 0.0005-inch diameter orifice for the high pressure air source of 4.4.3.3.

4.4.4 Test procedure. The following tests shall be performed:

WARNING

When working with high-pressure air or electrical apparatus, observe all safety precautions (see 3.2.7).

* 4.4.4.1 Hydrostatic pressure tests

* 4.4.4.1.1 Breech mechanism. With the breech mechanism removed from the tube assembly, remove Gage 1623922 and insert a standard 1/8 pipe plug in the void. The breech mechanism shall be filled with water and pressurized

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:41Z
Check the source to verify that this is the currentversion before use.

to 4500 plus 20 minus 0 psig. This pressure shall be maintained for 10 minutes minimum without failure of any part. At the completion of this test, reduce the pressure to zero, drain the breech mechanism, and purge with dry nitrogen. Reinstall Pressure Gage 1623922.

* 4.4.4.1.2 Low-pressure pneumatic system. Fill the low-pressure pneumatic system with water and pressurize it to 225 psig. Maintain this pressure for 10 minutes with the barrel in the SAFE condition and for 10 minutes with the barrel in the READY condition. There shall be no failure of any part of the pneumatic system. After completion of the test, thoroughly purge water from the system with dry nitrogen.

* 4.4.4.2 High-pressure pneumatic system pressure drop test. Remove the breech mechanism and securing mechanism from the torpedo tube. Secure the nipple attached to the manifold and piping assembly to a shutoff valve or other suitable blanking device. Adjust the air source to 2000 plus 0 minus 20 psig and attach it through a 0.024 plus or minus 0.0005 inch diameter orifice to the barrel pneumatic system using a Pressure Fitting Adapter (Dwg 3245120). Actuate the firing solenoid valve and allow the pneumatic system line pressure to stabilize. The pressure as measured on the barrel side of the orifice shall stabilize at 1350 psig minimum. Repeat this test with the securing mechanism installed in the barrel. The pressure in the pneumatic system shall then stabilize at 750 psig minimum. At the conclusion of this test there shall be no deformation or failure of any part of the pneumatic system.

4.4.4.3 Breech mechanism air loss. Remove the breech mechanism from the tube assembly and, using the test equipment of 4.4.3.3, 4.4.3.4, and 4.4.3.5, gradually charge the mechanism at a rate not exceeding 250 psig plus or minus 50 psi per minute to a pressure of 2000 psig pneumatic. Permit the mechanism to cool for a period of 2 hours. The breech mechanism shall then be bled to a pressure of 1600 plus or minus 10 psig. After a 24-hour period, observe the pressure reading, using the gage of 4.4.3.5. The pressure loss in the breech mechanism shall not exceed 5 percent. Compensation may be made for change in pressure due to temperature.

* 4.4.4.4 Bore size of barrel. The "GO" bore gage 4.4.3.6 shall pass freely through each barrel with an axial force not to exceed 150 pounds.

* 4.4.4.5 Dielectric strength. For the purpose of this test, the barrel heater switch shall be placed in the ON position and the barrel heater thermoswitches shall be cooled with CO_2 to close the contacts. The tube assembly shall withstand a potential of 1000 volts, 60 Hz ac (4.4.3.7) applied between terminal 14 of TB103 (in the junction box) and all other input terminals as shown on the External Electrical Schematic, Dwg 3014877, for one minute without evidence of breakdown. Input terminals may be connected together for this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

4.4.4.6 Insulation resistance. Immediately after the dielectric strength test, the insulation resistance between hipotted points shall be 3 megohms minimum when a potential of 500 volts dc is applied.

* 4.4.4.7 Ground circuit continuity. Continuity shall exist between terminal 14 of TB103, pin G of test receptacle J2 (in the junction box), and any portion of the junction box as well as the control box and overheat sensor assembly.

* 4.4.4.8 Barrel heater resistance test. With the barrel heater switch in the OFF position and the leads from the junction box to terminals 3 and 4 of TB101 in the control box disconnected, cool the two thermoswitches with CO_2 to close the contacts. The resistance between terminals 3 and 4 of TB101 shall be 11.4 ohms plus or minus 0.8 ohm.

* 4.4.4.9 Heater circuit operating currents. For the purpose of this test, the 125 volt dc power supply of 4.4.3.23 shall be applied to terminals 5 and 6 of TB103. With the thermoswitches in the barrel cooled with CO_2 to close the contacts, with 115 volts ac plus or minus 0.5 volt ac applied to terminals 3 and 7 to TB103 in the junction box, and with an ammeter installed in the circuit switch and the barrel heater switch in the junction box set to the ON position, the current in the circuit shall not exceed 11 amperes.

4.4.4.10 Squib fire switch. With the control box latch plate in the locked position, continuity shall exist across terminals 8 and 9 of TB101 and no continuity shall exist across terminals 7 and 8 and terminals 7 and 9 of TB101. Release the latch on the control box, observing that the switch plunger moves out forcefully, and verify that continuity exists across terminals 7 and 8 of TB101 and that no continuity exists across terminals 8 and 9 and terminals 7 and 9 of TB101.

4.4.4.11 Muzzle cover test. With the muzzle cover installed, a pull scale attached to the center of the cover handle, and a pulling force exerted in the direction perpendicular to the plane of the cover, the axial force required to remove each muzzle cover shall be 30 plus or minus 10 pounds.

4.4.4.12 Breakaway plug puller mechanism. Verify that the lever of the breakaway plug puller mechanism can be latched and unlatched by hand without excessive force.

4.4.4.13 Pressure switch operation

* 4.4.4.13.1 Utilizing a suitable means to indicate electrical continuity, gradually apply pressure to the air flask at a rate not exceeding 250 psig plus or minus 50 psig per minute until the pressure switch closes. The pressure switch shall close at 1275 plus or minus 75 psig, and remain closed at pressures above this value as indicated by a continuity reading.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:41Z
Check the source to verify that this is the current version before use.

* 4.4.4.13.2 Gradually decrease the pressure until the pressure switch opens. The pressure switch shall open at 1200 psig minimum and remain open at pressures below this value.

* 4.4.4.14 Dry firing tests. The breech mechanism shall be charged gradually, at a rate not exceeding 250 psig plus or minus 50 psig per minute, to the pressures specified in Schedule I. With the control valve operating lever in the READY position, the plug puller mechanism in the unlatched position, the breakaway plug disconnected from the plug puller mechanism and secured to the guide, and all firing interlocks satisfied, the barrel shall be fired a total of six times without a torpedo in place. Four of the firings shall be made by electrical means and two by mechanical means. The power supply of 4.4.3.10 shall be varied as specified in Schedule I for the four electrical firings. The application of voltage shall be regulated by the test circuit of 4.4.3.16, which shall limit firing potential duration to 400 milliseconds maximum.

* 4.4.4.14.1 Electrical firing. The barrel electrically fired shall fire with only one 400-millisecond application of test potential. Two firings shall be made at 102.5 plus or minus 0.5 volts ac and two at 123.5 plus or minus 0.5 volts ac.

4.4.4.14.2 Manual firing. During the manual firing tests measure the force required to activate the solenoid valve with the push scale. The force required shall be 25 pounds maximum.

4.4.4.14.3 Breakaway plug mechanism. After each dry firing, the plug puller mechanism shall be in the latched position and shall be capable of being unlatched by hand.

4.4.4.14.4 Control valve operating lever. During all dry firing tests verify that there is no leakage or premature air escape while the control valve operating lever is in the STANDBY condition. While performing the tests of Schedule I, with the control valve operating lever in the STANDBY condition, attempt to manually and electrically fire the tube. The tube shall not fire and there shall be no loss of pressure in the breech mechanism.

4.4.4.14.5 Pressure switch operation. During the dry firing tests verify that the pressure switch condition is in accordance with Schedule I when the control valve operating lever is in the READY condition.

4.4.4.14.6 Breakaway plug mechanism. The breakaway plug mechanism shall permit firing when attached in the breechward position and shall remain in the latched position at the completion of the firing.

* 4.4.4.14.7 Observation for damage. At the completion of the dry firing tests there shall be no damage sustained by the torpedo tube assembly.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:41Z
Check the source to verify that this is the current version before use.

SCHEDULE I

DRY FIRING TESTS

No. of Firing	Flask Pressure	AC Input Voltage to Solenoid	Pressure Switch Condition	Fire Mode
1	1200 psig	-	-	Manual
2	2000 psig	-	-	Manual
3	1350 psig	102.5 ± 0.5	Closed	Electrical
4	2000 psig	102.5 ± 0.5	Closed	Electrical
5	1350 psig	123.5 ± 0.5	Closed	Electrical
6	2000 psig	123.5 ± 0.5	Closed	Electrical

NOTE: Prior to conducting the above tests remove the clevis pin from the lever of the plug puller mechanism and disconnect the lanyard from the lever. Reinstall and secure the clevis pin through the keeper and lever. Secure the breakaway plug to the guide by looping the lanyard over the plug.

4.4.4.15 Continuity. Continuity shall be in accordance with the applicable schematic diagram.

4.4.4.16 Lanyard retainer. The lanyard retainer plunger shall be depressed by hand and a loop of 0.069 inch minimum diameter wire shall be assembled to the retainer.

4.4.4.16.1 The plunger shall depress easily by hand pressure.

4.4.4.16.2 The retainers shall accept and secure the loop of 0.069 inch minimum diameter wire.

* 4.4.4.17 Torpedo exit velocity tests. The two photoelectric control units (4.4.3.20) shall be spaced 3 feet apart alongside the track or ramp. The first set of cells shall be located the length of the torpedo plus or minus 0.5 foot from the muzzle end of the barrel. The photoelectric cells shall be attached to the electronic counter (4.4.3.21) so that when the first beam is broken by the nose of the torpedo the counter starts and when the second beam is broken by the nose of the torpedo the counter stops. The average velocity shall then be calculated by the formula $V = d/T$ where:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

d= distance in feet

V= average velocity in ft/second

T= time in seconds

The breech mechanism shall be charged gradually at a rate not to exceed 250 psig plus or minus 50 psig per minute to pressures in accordance with Schedule II. The minimum exit velocity shall be met in all firings.

* SCHEDULE II

TORPEDO EXIT VELOCITY TESTS

No. of Launchings	Torpedo Type (Mk)	(Mod)	Air Flask Pressure (psig)	Pressure Switch Condition	Firing Method	Minimum Exit Velocity (ft/sec)
1	44	1	1200	-	Manual	40
1	44	1	1350	Closed	Electrical	40
1	44	1	1500	Closed	Electrical	40
1	46	1	1200	-	Manual	40
1	46	1	1350	Closed	Electrical	40
1	46	1	1500	Closed	Electrical	40

NOTE: Should the velocity test methods of 4.4.4.17 be impractical or difficult for the manufacturer, an alternate means of measuring exit velocity, proposed by the manufacturer and approved by the contracting activity, shall be considered a valid test.

* 4.4.4.17.1 Electrical firing test. Each electrical firing test shall be made with only one 400 millisecond application of test potential. The test potential used for these tests shall be 115 vac plus or minus 0.5 vac.

4.4.4.17.2 Manual firing. During the manual firing test, measure with the push scale the force required to actuate the solenoid valve. The force required shall be 25 pounds maximum.

* 4.4.4.17.3 Breakaway plug mechanism.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

* 4.4.4.17.3.1 Prior to each firing, the breakaway plug mechanism lever shall be placed in the unlatched position and the breakaway plug attached to the torpedo receptacle.

* 4.4.4.17.3.2 After each firing, the plug puller mechanism lever shall be in the latched position and shall be capable of being unlatched by hand.

* 4.4.4.17.3.3 After each firing, the breakaway plug shall be inspected for damage.

4.4.4.17.4 Pressure switch operation. During the firing tests, verify that continuity of the pressure switch circuit exists in accordance with Schedule II when the control valve operating lever is in the READY condition.

4.4.4.17.5 Control valve operating lever. During all tests, observe that when the control valve operating lever is in the STANDBY condition there is no leakage or premature air escape.

4.4.4.17.6 Ready cylinder assembly solenoid valve operation. The ready cylinder assembly shall be operated electrically in accordance with Schedule II. Proper operation shall be indicated by control valve lever operation.

4.4.4.17.7 Ready cylinder assembly operation. The ready cylinder assembly shall operate in such a manner that the control valve operating plunger shall be positioned in the READY position when the solenoid valve is energized with 115 vac.

4.4.4.17.8 Observation for damage. At the completion of the firing tests there shall be no damage to the barrel assembly.

4.4.4.18 Visual and dimensional examination. Each torpedo tube shall be examined to determine conformance with the requirements for workmanship and dimensions.

4.4.4.19 Packaging, packing, and marking. Examine the packaging, packing, and marking to determine conformance with the requirements for workmanship and dimensions.

4.5 Preproduction and periodic production tests. Each preproduction and periodic production sample, after satisfactorily passing the quality conformance tests and examinations of 4.4, shall be subjected to the following environmental tests. Unless otherwise specified, the sample shall be deemed defective if it fails to satisfactorily pass the quality conformance tests and examinations subsequent to these environmental tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

4.5.1 Vibration tests

4.5.1.1 Vibration, nonoperating. The torpedo tube, secured to the mounting bracket of the testing machine in the same manner that it will be secured on shipboard, shall be subjected to the exploratory vibration test, variable frequency test, and endurance test specified in MIL-STD-167 for Type I vibrations.

4.5.1.2 Vibration, standby condition. The barrel of the torpedo tube shall be loaded with a dummy Mk 46 Torpedo. The breech assembly shall be pressurized to 1500 psig. The torpedo tube, loaded as specified and secured to the mounting bracket of the testing machine in the same manner that it will be secured on shipboard, shall be subjected to the exploratory vibration test, variable frequency test, and endurance test specified in MIL-STD-167 for Type I vibrations.

4.5.2 Shock tests

* 4.5.2.1 Shock, nonoperating. The torpedo tube shall be attached to the anvil table of the shock testing machine by means of a standard mounting platform and a 30-degree mounting fixture as specified in MIL-S-901 for medium-weight equipment. The torpedo tube shall be oriented on the 30-degree mounting fixture such that the barrel centerline is parallel to the horizontal plane. In each mounting condition the torpedo tube shall be subjected to Type A shock tests as specified in MIL-S-901 for medium-weight equipment.

* 4.5.2.2 Shock, loaded condition. The torpedo tube shall be attached to the anvil table of the shock testing machine by means of a standard mounting platform and a 30-degree mounting fixture as specified in MIL-S-901 for medium-weight equipment. The torpedo tube shall be loaded with a dummy torpedo Mk 46 Mod 1. The torpedo tube shall be oriented on a 30-degree mounting fixture such that the barrel centerline is 30 degrees from the horizontal plane. In each mounting condition the torpedo tube shall be subjected to Type A shock tests as specified in MIL-S-901 for medium-weight equipment.

4.5.3 Temperature and humidity tests

4.5.3.1 Low temperature, loaded condition. The torpedo tube shall be loaded with a dummy Mk 46 Torpedo. Thermocouples shall be installed on the surface of the torpedo in the area of contact between the torpedo and the barrel quadrant. Provisions shall be made for reading the sensed temperature. The torpedo tube shall be placed in a test chamber and the barrel heaters actuated with 115 plus or minus 0.5 volts ac. The 125 volt dc power source of 4.4.3.23 will be required to activate the heater cutout relay. The temperature of the test chamber shall be lowered to -29 $\pm$ 3 degrees C (-20 $\pm$ 5 degrees F) and maintained at this temperature for a period of 48 plus or minus 5 hours. At the end of 48 plus or minus 5 hours, the temperature shall be recorded and the complete torpedo tube shall be performance checked. The temperature recorded in the battery compartment shall at no time exceed 37.5°C (100°F) or be less than 0.5°C (33°F).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

coils, marking of parts and assemblies, plating, riveting, machine screw

* 4.5.3.2 High temperature and humidity tests. The torpedo tube assembly, with fully pressurized breech assembly (2000 psig), shall be placed within a test chamber and subjected to temperature and humidity cycling as specified in MIL-STD-810, Method 507, Procedure IV, with the following exceptions:

(a) Referenced measurements of steps 4,5,6,and 8 may be deleted.

(b) The 24-hour cycles of step 5 shall have a high-temperature minimum of 59°C (138°F) and a maximum as specified.

(c) The torpedo tube shall be performance checked during the high temperature portion of the fifth cycle of step 5 prior to decreasing to 30°C (86°F).

(d) Energize the heater cutout relay using the power supply of 4.4.3.23 and monitor the continuity between terminals 3 and 4 of the relay. Continuity shall exist between these terminals during the low-temperature phase of the cycles and no continuity shall exist during the high-temperature phase of the cycles.

4.5.4 Marine environment

4.5.4.1 Marine atmosphere environment test. The torpedo tube assembly shall be exposed at dockside or on the weather deck of a ship for 60 plus or minus 5 days. The access cover and muzzle cover shall be removed for a period of 10 plus or minus 1 days. The assembly shall be protected from rain or snow by an appropriate shelter that will permit free circulation of air. At the completion of this test, there shall be no evidence of corrosion.

* 4.5.4.2 Alternative test for marine atmosphere. Subject the torpedo tube assembly to the Salt Fog test of MIL-STD-810, Method 509, using a 20 percent salt solution. The solution shall be prepared by dissolving 20 plus or minus 2 parts by weight of salt in 80 parts by weight of distilled water. The solution shall be adjusted to and maintained at a specific gravity between 1.126 and 1.157. At the completion of this test there shall be no evidence of corrosion.

4.5.5 During and after performance of the above environmental tests, no damage shall occur which could affect proper operation or function of the tube.

4.5.6 At the completion of the above environmental tests, the torpedo tube shall meet all the quality conformance tests of 4.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C, as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Cleaning. Unless otherwise specified, the torpedo tube, in the nonoperating condition, shall be cleaned in accordance with Method C-1 of MIL-P-116.

5.1.1.2 Drying. Unless otherwise specified, the torpedo tube shall be dried in accordance with one or more of the procedures of MIL-P-116, as appropriate.

5.1.1.3 Preservation. Immediately after cleaning and drying the torpedo tube, the external unpainted metal surfaces which are subject to corrosion shall be preserved by the application of preservative compound conforming to the requirements of MIL-C-16173, Grade 2.

5.1.1.4 Unit packaging. Not applicable.

5.1.2 Level B. Not applicable.

5.1.3 Level C. Cleaning, drying, preservation, and packaging shall be in accordance with the supplier's commercial practice when such practices have been approved by the procuring activity.

5.2 Packing

5.2.1 Level A

5.2.1.1 Exterior containers. The torpedo tube assembly, in the nonoperating condition, shall be packed in a nailed wood, skid-mounted crate with rubbing strips in accordance with MIL-C-104 for Class 1, Style A, Type I crates. The torpedo tube saddles shall be rigidly fastened to the load bearing members of the skid base and the tube barrel shall be braced and cushioned to prevent movement.

5.2.2 Level B. Not applicable.

5.2.3 Level C. The torpedo tube assembly, in the nonoperating condition, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall comply with the carrier rules and regulations applicable to the mode of transportation.

5.3 Marking

5.3.1 Special markings. None unless otherwise specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

5.3.2 Normal markings. In addition to the markings required by contract or order, shipping containers shall be marked in accordance with MIL-STD-129, including serial numbers.

* 5.4 Inspection data sheets. When Quality Conformance Inspection Data sheets are specified in the contract or order (See 6.2.2) for completion in accordance with 4.4.2, one data sheet (or copy thereof), bearing recorded data, shall be shipped with each torpedo tube assembly inspected. The torpedo tube assembly equipment serial number shall correspond to the serial number recorded on the accompanying data sheet.

6. NOTES

6.1 Intended use. Torpedo Tube assembly Mk 32 Mod 11 is used on the weather decks of Naval surface vessels to launch Torpedoes Mk 44 and Mk 46. Torpedo Tube Assembly Mk 32 Mod 11 is intended for use with Control Panel Mk 264 or Mk 309 of Fire Control Systems Mk 102, Mk 105 (except Mod 24), and Mk 111.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) The specific number of tubes being procured.

(c) Whether a Quality Conformance Inspection Data Sheet is required (see 4.4.2).

(d) Required level of preservation and packaging (see 5.1.1 and 5.1.3).

(e) Whether preproduction tests apply, the assigned test activity, and instructions for submittal of the preproduction sample (see 3.2.1 and 4.3.1).

(f) Whether periodic production tests apply, the assigned test activity, and instructions for submittal of the periodic production samples (see 4.3.3).

(g) Any exception to this specification or its applicable drawings and documents.

(h) That all additional samples required shall be provided and inspected at the expense of the supplier, when such additional samples are required because of failure of any sample(s) to pass the prescribed inspections.

6.2.1 The attention of the contracting officer is invited to the Quality Assurance Provisions and options in MIL-P-116.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

Fire Control (FC) Switchboard

* 6.2.2 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraph herein:

Paragraph	Data Requirement	Applicable DIDs
4.4.2	Test Results	UDI-E-20602

DIDs (Data Item Descriptions/DD Forms 1664) for the above requirements will be documented in the applicable ADL (Authorized Data List). Such data will be delivered as identified on completed (numbered) DIDs when specified on DD Forms 1423 (Contract Data Requirements Lists) and incorporated into applicable contracts.

6.3 Definitions. For the purpose of this specification the following definitions are applicable:

6.3.1 Torpedo tube assembly. The launching device in its assembled state.

6.3.2 Nonoperating condition. The condition of the tube when assembled, prepared, issued, transported, and handled.

6.3.3 Loaded condition. That condition when a torpedo or dummy is installed in the tube barrel, secured in place by the securing mechanism, and the breech mechanism and muzzle covers are installed.

6.3.4 Latched position of the breakaway plug puller mechanism. The breakaway plug puller mechanism is latched when the lever is in the breechward position.

* 6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian
Navy - OS

Preparing Activity
Navy - OS
Project No. 1045-N009

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

DATA SHEET

QUALITY CONFORMANCE INSPECTION

TESTS AND EXAMINATIONS

FOR

TORPEDO TUBE MARK 32 MOD 11

(LD 162748)

EQUIPMENT STATUS

CONTRACTOR CERTIFICATION		GOVERNMENT ACCEPTANCE	
Contractor		Date of Acceptance	
Contract		Accepted by	
Equipment Ser. No.		Title	
Insp. Req. Satisfied		Agency	

* ORDALT STATUS

ORDALT NO.	INSTALLED	ORDALT NO.	INSTALLED

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

DATA SHEET

QUALITY CONFORMANCE INSPECTION

TORPEDO TUBE MARK 32 MOD 11 Equipment Serial No. _____

10. SCOPE

10.1 This appendix contains data sheets to record the results of the Quality Conformance tests and examinations of 4.4 of this specification. When completed, with authorized certification and acceptance signatures affixed, this report form or copies thereof shall accompany the Torpedo Tube Assembly bearing the recorded equipment serial number.

20. GENERAL INSTRUCTIONS

20.1 Test sequence. When the contract or order specifies use of this appendix, the Quality Conformance Inspection tests of 4.4 shall be completed and the results recorded or verified in these data sheets prior to acceptance of the Torpedo Tube Assembly. Unless otherwise noted, the sequence of testing procedures presented herein is suggested only. The procedures may be performed in any order which will permit the most practical and economical use of manpower and equipment.

20.2 Recording of results. Where space is provided in the "Reading" column of the Quality Conformance Test data sheets, the value or specific reading obtained shall be recorded and the initials of the individual performing the inspection shall be inserted above the value observed and recorded. If limits are not stated, the individual performing the test or inspection shall initial in the space provided in the "Reading" column indicating that the requirements have been met satisfactorily.

Paragraph	Reference	Function	Remarks
30.1 Test Equipment	4.4.3	Calibration	_____
30.2 Test Procedure	4.4.4	Procedure	_____
30.2.1 Hydrostatic pressure test (breech mechanism)	4.4.4.1.1	Verified	_____
* 30.2.2 Hydrostatic pressure test (low pressure pneumatic system)	4.4.4.1.2	Verified	_____

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

	Paragraph	Reference	Function	Remarks
*	30.2.3 Pressure drop test (high pressure pneumatic system)	4.4.4.2		
	Test No.1		1350 psig min.	_____
	Test No. 2		750 psig min.	_____
	30.2.4 Breech mechanism air loss test	4.4.4.3	80.8 pounds maximum loss	_____
*	30.2.5 Bore size of barrel test	4.4.4.4	150 lbs max	_____
	30.2.6 Dielectric strength tests	4.4.4.5	No breakdown	_____
	30.2.7 Insulation resistance test	4.4.4.6	3 megohms minimum	_____
	30.2.8 Ground circuit continuity	4.4.4.7	Verified	_____
*	30.2.9 Barrel heater resistance test	4.4.4.8	11.4 ohms ±0.8 ohms	_____
*	30.2.10 Heater circuit operating current	4.4.4.9	11 amps max	_____
	30.2.11 Squib fire switch	4.4.4.10	Operation	_____
	30.2.12 Muzzle cover test	4.4.4.11	30 pounds ± 10 pounds	_____
	30.2.13 Breakaway plug puller mechanism	4.4.4.12	Verified operation	_____
	30.2.14 Pressure switch operation	4.4.4.13		
	Test No. 1		1275±75 psig	_____
	Test No. 2		1200 psig min	_____
	30.2.15 Dry firing tests	4.4.4.14	Verified	_____

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

	Paragraph	Reference	Function	Remarks
*	30.2.16 Electrical firing test	4.4.4.14.1		
	Test No. 1		102.5 VAC	_____
	Test No. 2		123.5 VAC	_____
	30.2.17 Manual firing	4.4.4.14.2	25 pounds max	_____
	30.2.18 Breakaway plug mechanism	4.4.4.14.3	Verified operation	_____
	30.2.19 Control valve operating lever	4.4.4.14.4	Verified operation	_____
	30.2.20 Pressure switch operation	4.4.4.14.5	Verified per Schedule I	_____
	30.2.21 Breakaway plug mechanism	4.4.4.14.6	Verified operation	_____
	30.2.22 Observation for damage	4.4.4.14.7	No damage	_____
	30.2.23 Continuity	4.4.4.15	Verified operation	_____
	30.2.24 Lanyard retainer	4.4.4.16	Verified operation	_____
	30.2.25 Torpedo exit velocity for Barrel Mk 10 MOD 19	4.4.4.17	Verify in accordance with Schedule III	_____

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:41Z
Check the source to verify that this is the current version before use.

SCHEDULE III

* TORPEDO EXIT VELOCITY TESTS

No. of Launchings	Torpedo Type (MK)	(MOD)	Air Flask Pressure (psig)	Pressure Switch Condition	Firing Method	Minimum Exit Velocity (ft/sec)	Measured Exit Velocity (ft/sec)
1	44	1	1200	-	Manual	40	
1	44	1	1350	Closed	Electrical	40	
1	44	1	1500	Closed	Electrical	40	
1	46	1	1200	-	Manual	40	
1	46	1	1350	Closed	Electrical	40	
1	46	1	1500	Closed	Electrical	40	

U.S. GOVERNMENT PRINTING OFFICE: 1975— 603-130:1351

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T04:41Z
Check the source to verify that this is the currentversion before use.

FOLD

POSTAGE AND FEES PAID

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commanding Officer
Naval Ordnance Station (502)
Standardization Division
Indian Head, Maryland 20640

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-T-82612A(OS) Torpedo Tube, Surface Vessel, Mark 32 Mod 11

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

☆ U.S. GOVERNMENT PRINTING OFFICE: 1972—714-863/3404 2-1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.